

Zaid Shaikh

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EXPERIENCE

PIVOTCHAIN SOLUTIONS

Oct 2024 – Present

Data Scientist

Pune, Maharashtra

- Designed an **NLP-driven resume shortlisting pipeline** using Transformer-based semantic embeddings and similarity scoring, processing over 1,200 candidate profiles across 3 hiring cohorts; reduced manual HR screening effort by 60% and cut average shortlisting time from 3 days to under 4-5 hours.
- Developed a real-time person tracking and face recognition system using a **ResNet-100** embedding backbone across a 6-camera deployment; applied knowledge distillation to **ResNet-18** and **ResNet-50** student models, compressing model size by **60%** and reducing inference latency by 30% while maintaining recognition accuracy above 94% on the validation set.
- Engineered a real-time minor detection and alerting system for wine shop CCTV surveillance using **YOLOv8** and **YOLOv11** detectors with a custom YOLO-based age classifier trained on **8,000 labeled samples**; achieved **>90% classification accuracy** and eliminated manual monitoring across 4 live camera feeds, cutting false-alert rate to under 5%.
- Built a real-time **ANPR pipeline** integrating multi-object tracking, **EasyOCR** and **PaddleOCR**, and OpenCV-based preprocessing; diagnosed and resolved 5 critical pipeline bottlenecks including frame-drop under high vehicle density, OCR misreads on low-contrast plates, and tracker ID switches achieving stable **15-20 FPS** inference with plate-read **accuracy above 92%** under varying lighting conditions.
- Built scalable backend infrastructure for enterprise AI products including a **MongoDB-based DAL** and **RabbitMQ** publisher system handling over 500 async events/min; integrated real-time video processing over RTSP and RTMP supporting **50+ concurrent streams** across production deployments.
- Maintained **end-to-end containerized deployment** workflows on Linux across 3+ production AI services, standardizing build pipelines and reducing environment-related deployment failures by 40%.

PROJECTS

Mini LLM – Decoder-Only Transformer Built from Scratch

- Designed and implemented a **20M parameter decoder-only transformer** entirely from scratch in PyTorch (no built-in transformer modules) and trained on an **RTX 4080 (8GB VRAM)**, covering BPE tokenizer, multi-head causal self-attention, positional encoding, feed-forward layers, and layer normalization.
- Pre-trained on **TinyStories (16M tokens)** achieving **val perplexity of 2.64** outperforming the typical 5-8 benchmark range for models of this size.
- Diagnosed and resolved **8 critical training bugs** including vanishing gradients across 6 decoder blocks, exploding activations through residual connections, attention score overflow.
- Performed **instruction fine-tuning** on 3,600 custom prompt-response pairs, enabling the model to generate coherent, on-topic children's stories from natural language prompts.
- Implemented full training infrastructure: checkpoint save/resume (model + optimizer + scheduler state), early stopping, cosine LR decay, and perplexity tracking per epoch.

Enterprise RAG System with Semantic Search and Citation-Based Question Answering

- Developed a real-time intelligent surveillance pipeline integrating **YOLO** for object-detection and multi-object tracking algorithms (DeepSort and Byte-Track) for automated person and vehicle monitoring from live CCTV streams.

Explainable Employee Attrition Prediction and Drift Monitoring

- Mentored 15 students towards acceptance at top US boarding schools; achieved **100% success rate**
- Designed a **collaborative learning ecosystem** for students and parents with Trello, Miro, and Google Suite

SKILLS

Programming & Development: Python, FastAPI, Flask

Machine Learning: Scikit-learn, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Explainable AI

Computer Vision: YOLO, OpenCV, Object Detection, Multi-Object Tracking, Face Recognition, ANPR,

Vector Databases: FAISS, Milvus, ChromaDB, Pinecone

Natural Language Processing: Sequence Modeling, Semantic Embeddings, Text Preprocessing, Resume Ranking Systems, BiLSTM, Transformer-based NLP Pipelines

Tools & Platforms: Linux, Git, Docker, ONNX Runtime, Pandas, NumPy, Jupyter Notebook

EDUCATION

APCOER, SPPU UNIVERSITY	2021 – 2024
<i>Bachelor of Engineering in Information Technology (Honours in Data Science)</i>	CGPA – 9.60/10
AISSMS POLYTECHNIC, MSBTE UNIVERSITY	2018 – 2021
<i>Diploma in Computer Engineering</i>	91.60 %
ANGLO BOYS HIGH SCHOOL	2018
<i>SSC BOARD 10TH</i>	81.40 %